

Mei-Yu Lai

Computational Biologist | Boston, MA | 443-467-2562 | meiyualai@gmail.com
Personal website: <https://mlai22.github.io> |  meiyulai |  mlai22 |  Google Scholar

SUMMARY

Computational biologist with 8+ years of experience applying machine learning, multi-omics integration, and systems biology approaches to identify therapeutic targets and uncover disease mechanisms. Deep expertise in functional genomics, single-cell transcriptomics, proteogenomics, and statistical modeling of large-scale datasets. Proven ability to lead cross-functional research, translate complex analytical insights into actionable biology, and contribute to drug discovery in oncology and neurodegeneration.

CORE SKILLS

ML & Statistics: predictive modeling, disease subtype stratification, feature selection, biomarker discovery, survival analysis, longitudinal progression analysis

Multi-Omics & Functional Genomics: multi-omics integration (Bulk RNA-seq, proteomics, transcriptomics, metabolomics), single-cell RNA-seq (scRNA-seq), GWAS, pQTL, colocalization, fine-mapping

Programming & Computing: Python, R, MATLAB, Shell scripting, HPC/Slurm, Git

Bioinformatics Tools: PLINK, STAR, variant prioritization

Communication: Technical presentations at international conferences; cross-functional collaboration with wet-lab and clinical teams

EXPERIENCE

Data Scientist

04/2022 – Present

Beth Israel Deaconess Medical Center, Boston, MA

- Develop end-to-end proteogenomic pipelines on HPC infrastructure integrating large-scale GWAS and pQTL data to identify and prioritize causal proteins
- Implement single-cell transcriptomic analyses that resolved cell-type-specific gene expression patterns underlying meningeal lymphatic dysfunction in PD
- Deliver multi-omics analytical frameworks to uncover novel biomarkers and built predictive machine learning models to stratify disease subtypes

Bioinformatician

06/2021 – 09/2021

Johns Hopkins University, Baltimore, MD

- Prioritized causal variants in rare disease in Critical Assessment of Genome Interpretation (CAGI6) challenge
- Tested and evaluated predictions of causal variants and interpreted data to clinical relevance

Biostatistician

08/2012 – 06/2016

Academia Sinica, Taiwan

- Designed and constructed an in-house website for sample size calculations in various designs
- Investigated differential expression genes in breast cancer and developed statistical models for prognosis of distant metastasis
- Analyzed clinical data to support development of clinical trial protocols

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PROJECTS

Proteogenomic insights of Parkinson's disease (PD)

Discovered causal PD proteins using systems biology approach that could be used as drug targets and offers a novel methodology to improve our understanding of PD pathogenesis and prediction.

Multi-omics approaches applied to neurodegenerative diseases

Used high-throughput omics technologies to uncover potential biomarkers that will improve our understanding of underlying disease mechanisms, predict disease causality, and improve the design of clinical trials.

Cellular Atlas of Parkinson's Disease

Uncovered cell-type-specific differentially expressed genes for deciphering the role of meningeal lymphatics in PD.

Gut-brain axis in Parkinson's disease

Identified signatures to distinguish PD patients at higher risk of developing severe gastrointestinal disease using a multi-omic approach, thereby benefiting personalized medicine.

BRCA1 gene expression driven metastatic risk in ER+ Breast cancer

Employed a logistic-AFT mixture regression model to elucidate the link of BRCA1 gene expression and aggressive tumor behavior.

Longitudinal data analysis for metabolic profiles in polycystic ovary syndrome

Utilized generalized estimating equation in a segmented regression model to assess the heterogeneity in response to treatment duration.

EDUCATION

MSE, Biomedical Engineering (Genomics & Systems Biology)

12/2021

Johns Hopkins University, Baltimore, MD

MS, Biometry

06/2012

National Taiwan University, Taiwan

BS, Applied Mathematics

06/2010

Chung Yuan Christian University, Taiwan

SELECTED PRESENTATIONS

Digestive Disease Week (Talk), Washington DC

05/2024

Lai MY, Kulkarni S, and Benitez BA. Molecular Markers Associated with Self-reported Progressive Gut Dysfunction in Parkinson's Disease Patients.

American Academy of Neurology Annual Meeting (Talk), Boston, MA

04/2023

Lai MY, Palanisamy D, and Benitez BA. Proteomic Signatures Associated with Genetic and Clinical Subtypes of Parkinson's Disease.